

# Asif Iqbal Rahaman

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PhD fellow working with Phylo-Oncology and ML in Genomics in Suvorov Lab at Virginia Tech. Computational Biologist and Software Engineer with expertise in bioinformatics, machine learning, and cloud infrastructure. Research interests include integrating machine learning and optimization methods into genomic data analysis and algorithm development.

## Key Skills

Python, R, Rust, Pytest, Docker, PyTorch, Matplotlib, AWS, Azure, Linux, Spark, Pandas, ReactJS, Bash, SQL, NoSQL, Git, GitHub Actions, Tableau, C++, FastAPI, Flask, RestAPI, TypeScript, Microservices, OpenCV, TensorFlow, Jenkins, Kubernetes, Terraform

## Education

Virginia Tech, Blacksburg, VA

Aug 2025 – Present

- **Doctor of Philosophy (PhD)** in Biological Science (Bioinformatics) CGPA: 4.0
- **Projects:** Ghost Introgression detection, Cancer Cell Analysis from Evolutionary Lens, Phylogenetic Inference using ML

Virginia Tech, Blacksburg, VA

Aug 2022 – May 2024

- **Master of Engineering** in Computer Science (Data Science and AI) CGPA: 3.9

Vellore Institute of Technology, Vellore, India

Aug 2015 – May 2019

- **Bachelor of Technology** in Computer Science (Bioinformatics) CGPA: 3.4

## Experiences

Software Developer, Department of System Biology, Virginia Tech

Oct 2022 – Jul 2025

- Formulated **novel algorithm** for cell cycle perturbation with **large network**, enabling accurate **pathway prediction** and **data analysis** for targeting cancerous cells; published as joint first author (**computational lead**) in PLOS One.
- Implemented **stochastic simulation** with **vectorization**, **CPU** and **GPU** parallelization for **5x performance gain**.
- Developed **modular system** with **CLI parameter** support to be executed in **ARC HPC** cluster for **50%** efficiency gain.
- Improved reliability by **30%** using **data validation** pipeline from SIGNOR database and **visualization** using networkx.

**Skills:** Python, Pandas, Numpy, JAX, PySpark, Machine Learning, Parallel Processing, Linux, Database, Distributed Systems

Product Engineer, Seclore Technology Pvt Ltd

Dec 2021 – Jul 2022

- Automated customer onboarding for security tool (**90%** speedup) using **AWS CDK**, collaborating with **DevOps** and **QA**.
- Orchestrated integration of **AWS** services to build **IaC** with Agile **SDLC**, contributed to features with **70%** improvement.
- Built **CI/CD pipelines** for containerized layered architecture; applied **test-driven development** with **80%** coverage.

**Skills:** Python, AWS-CDK, Docker, Boto3, Jenkins, Lambda, DynamoDB, ECS, Terraform, Kubernetes, Agile Development

Software Engineer, Ericsson Global Services Pvt Ltd

Jan 2019 – Jul 2021

- Developed **backend** for custom **RPA framework** automating network tasks with **35%+** efficiency and **80%+** reliability.
- Built a rule-based **recommendation engine** for **big data analysis** (from network cells) with over **50%** automation gain.
- Architected **scalable ETL pipeline** for data processing and storing (~**50GB** parquet daily) with **CI/CD pipeline**.
- Received **biannual Galactic Award** for developing multiple **automation frameworks** with **50%** improvement in **2020**.

**Skills:** Python, PySpark, Flask, FastAPI, OpenCV, Linux, MySQL, PostgreSQL, MongoDB, PyTorch, Scrum, Apache Airflow

## Projects

**GhostParser: an efficient summary-statistics based Ghost Introgression detection tool**

Aug 2026

- Used summary-statistics like discordant count test, Kolmogorov Smirnov test, and perturbation tests on gene trees.
- Utilized average tree heights to reduce effect of gene recombination and reduce false positives by about **20%**.
- Optimized via parallel processing and vectorization to run **85-taxa** analysis with **2000 gene trees** within **2-3 hours**.

**Real-time Sentiment and Subjectivity Trend Analysis for Social Media**

Feb 2025

- Automated post ingestion (**APIs**, **GitHub Actions**, **CI/CD**); processed sentiment via **PySpark** on **Databricks** in **Azure**.
- Set up interactive **Streamlit** dashboard in **Azure** (via **GitHub Actions**) with real-time monitoring and visualization.

**Soccer Commentary/ Summary Generation with Assistant-based GPT APIs**

May 2024

- Built AI pipeline with **GPT APIs**, **prompt engineering** for event-driven, multilingual summary for real-time adaptation.
- Engineered frontend with **ReactJS**, integrating text, audio (**TTS**), and images (**DALL.E**) for multimodal experience.

For more details and other projects, please visit my [website](#) or [GitHub](#).

## Certifications

**Python for Data Science and Machine Learning Bootcamp**

Udemy (May 2021)

**Improving Deep Neural Networks: Hyperparameter Tuning, Regularization**

Coursera (Jul 2020)

**Neural Networks and Deep Learning**

Coursera (Jan 2020)